

Plotting with Different Data Structures Transcript

Also, I am also importing the bouquet output notebook and bouquet plotting. So and I am in this case trying to import bouquet models from column data source. Whenever you are trying to work with column data source, you have to import first the column data source.

So we are executing this command from bouquet models import column data source. So now we will be using the basic dictionary in Python to create a column data source. What is a dictionary? Dictionary is nothing but an ordered list.

So you have a column and the sequence of values that the column takes and then you have other column which are the other values. In this case, you have a variable x or column name x with values 1, 2, 3, 4, 5. Then you have a column y which has values 3, 7, 8, 5, 1. These are the two values or two columns that you have and you want to integrate these two columns into a column data source, two dictionaries into a column data source. The structure highlighted now is called a dictionary which is nothing but an ordered list indicating the sequence in which the elements will appear in the particular column or in the particular list.

So you can just take this command. Column data source is the function where it is saying the data is x, y. You can give the sequence of columns with the following values that it will be taking. So now that we have the column data source, let us see how to call the data from particular the data that is contained in column data source.

You can simply use the source is the variable name that you gave to this particular column data source and then source.data and you are typing column x. So once you hit this, run this, you get the entire column of the variable x. Likewise, you can also do y and call the data. This is similar to the subsetting of data that you do in a typical data frame. So now let us try to use the column data source.

You can also do this to NumPy, Arrays and Pandas series. We just did with the dictionary. We are now doing the NumPy, Arrays and Pandas series, so that which are string of values from which you are trying to create a column data source.

So let us first create a Pandas data series and generating random numbers. These are random integers, NumPy random integer generation function. I need 100 to 500 and I need the series, it is a 12 item series, there are 12 members in that particular series and there are random numbers between 100 to 500 and what do they indicate? They indicate monthly passengers in a year.

So I am just creating this series. I have created it. So it is a very simple command.

Then I am doing range. I want to create the NumPy array, which indicates the month values. So I am indicating it from let us say 0 to 11, indicating 12 months.

So range starts from 0 and then I have length of monthly passenger series. So how many numbers I want? I want numbers starting from 0 and plus 1 and the maximum length that the array can take is equal to the series, number of passengers that are created in the Pandas series, number of months that are in the Pandas series. Remember, the main requirement for column data source is to indicate, the main requirement for column data source is the number of elements in each column should be equal.

That's why we are saying that the range array should be equal to the monthly passenger series length. Okay. Same dimensions as the monthly passenger series.

Now we have created it. Now we'll be using the NumPy array and the NumPy series, Pandas series to concatenate them or join them in the column data source. So we are just collecting these two variables and creating a column data source.

So it's nothing but trying to say X variable is monthly passenger series. The first dictionary using the Pandas series, then Y is the range array, which is the second dictionary, which uses the NumPy array. So again, the same function source is equal to column data source is the function data is coming from X comma Y. Then you are just generating it.

Now let us see how does this series look like and just printing this out using the simple function print Pandas series source data X source data Y. So we are just getting this. So zero to 11. So it has generated zero to 11.

Okay. Within the length of the series and then so it starts from zero and it ends with the length of the monthly passenger series, which is 12 months. Okay.

And it increments by step one. That's how it is getting zero to 11. Then you have the number of passengers, which are random integers between say 100 to 500 each coming in the series.

Okay. This is how you can create a column data source along with the column data source using the dictionaries, lists, series from the Pandas and the NumPy framework. Always remember all columns in the column data source should be of same length.

So please remember this. Then now let us see how can we use, right, how can we use the normal functions. Okay.

So let us now see how to plot a scatter plot using two different methods in the column data source. So there are two typical ways in which you pass individual columns or data points into column data source directly. Okay.

You can pass column data source as a direct source into the glyph method, or you can use specific columns or call out specific columns from the column data source into the graph or the glyph as it called in the bouquet environment to achieve the figure that you are trying to plot. So let us simply look how to plot a scatter plot in the bouquet environment using these two methods. So I have imported the figure and show and also the column data source.

So I'm creating a simple data dictionary of X values and Y values. Okay. And just creating them.

So I have X values and Y values, two columns are created with the following values, and I'm creating a column data source with parsing the data from the data dictionary. Okay. So I have now also created the column data source.

These are ready. Now let us see how does calling the data works. So in this case, you are first setting out the figure, that is what you have to do in Matplotlib or in any environment.

First set out the figure function with height 300, and then it is p.scatter. Now on the p, the figure you are imposing or calling for the scatter function, trying to plot the scatter plot. X is equal to X values, Y is equal to Y values, which will come from these two, which is coming from the data dictionary, which is now converted as column data source. And source is, the data source is the variable called source.

So now I'm plotting this and finally you have to show p. So remember, in Bokeh, you have to use the function show to create the plot. So you are getting the plot now. Okay.

So it's very simple. Now let us see how to create column data source from a data frame. So like we have created the elements of column data source using the NumPy series or a dictionary or a list.

It is in a similar fashion, you would create a data frame and then parse it into column data source. But however, it is very simple compared to the other two methods we have seen earlier. First we create a sample data frame, we just concatenate two lists, months January to December, and data in those other months is passengers, rate and mail.

And these are random integers with the size 12 random integers for each of the columns and ranging between different numbers as given in the code. So now I'm just

creating this particular data frame so that you will be able to parse it into the column data source and use it for plotting the Bokeh Neurone. So it is very simple.

Passengers are random integers between 1000 and 5000. And freight is a random integer between 500 to 2000. Mail is a random integer between 100 to 500.

And each of them is size 12, which is there are 12 values in each of the series. And we are combining these four series to create a data frame with month name passengers freight and mail. Now to create a column data source from these, all that you have to do is use the simple command of using source is equal to column data source and parse the data frame in the brackets.

So you will now generated a column data source named source with the monthly values coming from this particular data frame. Likewise, if you have any readymade data frame, you can parse that into the column data source and create the corresponding column data source. Now let us see now that you have the column data source with same columns as the data frame.

How can you use the column data source for creating different figures? Okay, so I'm first using the I'm also trying to import colour palettes from the bokeh environment, which we have not used. So always remember, if a particular function is being used, it has to be imported into the Colab or into the Jupyter workbook, then only these functions will work. Otherwise, you get an error saying that a particular file is not there or the syntax error will come.

So just be mindful of what functions we are using. So source is column data source monthly values DF, which we already have created here. So then I'm trying to first determine the figure.

So I'm determining or calling the figure function and labelling it as P and width is 750. And x range, the x axis range, okay, sorry, x range is indicating the categorical value saying that the figure has categorical data, which means the x axis range is not continuous, it is categorical. And see, I'm using source dot data, which means I'm subsetting from the source column data frame, and then calling the month name and defining figure P. Now next, what I'm trying to do, I'm saying what are the columns, okay, I'm saying there are three columns, I'm labelling them as with a variable called columns.

And I have defined them freight passengers and mail, which is which are the columns in the data frame. Now, what am I doing, I'm creating a vertical bar, okay, using the bouquet. And also I'm also adding a legend and finally trying to show that in the diagram.

So let us read the code, `P dot v bar stack`, it's a stacked bar that we want. We don't want the normal vertical bar, we want to stack vertical bar the code for that is `v bar underscore stack`, then you are saying the columns are coming from the columns of the data column coming from the variable columns, freight passenger made, then `x` is equal to month name, width is equal to 0.9, source is the source date, source column data frame colours, we are using high contrast three, which we have just imported, and legend is equal to columns legend underscore label is equal to columns based on the column, which are again freight, you will have different legend and legend location is on the top left, you can also do right centre, okay, like that you can over it, and we have created this. So now this is a very good looking graph that is generated.

So in a January month, you have total is 3000 of which you have the passenger freight passengers and make in this order you give. So for example, if you are given freight passengers mail is the order in which you give the columns, if you reorder them automatically, this would get reordered as mail passengers and freight and correspondingly it will get generated. So this is how you can use a data frame to create a column data source and use it for plotting in the bouquet.

Okay, now let us understand column data source transforms, which are very interesting, useful tools that will help you to change the or apply certain numerical transformations on the underlying data for creating particular features in the data visualisation. But the point here is these transforms are performed by the bouquet JS in the browser, which means the underlying data is not modified and always available for other plots in the same document. So this is like always you take the source data, do some transformations and plot it, but not disturb the original data so that you can always use that for creating other plots without worrying about the transformations that you have applied in the earlier steps.

So now what we are trying to do is like we are trying to plot the cumulative sum cum sum transformation, which is cumulative summation of sequential values in a particular column. Let us say your column has 100 values, cumulative sum starts with first value, then second value is first value plus second value, third value is first value, second value plus third value. Likewise all if you go to the 100th value, it will be sum of all the 100 elements that are there in the column.

So cumulative sum is especially useful if you are seeing both the monthly sales and the overall sales at that particular point or especially in a modelling environment, the frequency distribution and cumulative frequency distribution, which can be converted into conditional probability and that can be used for predicting defaults. If you are talking in finance modelling, this can be a very good useful tool for creating visualisations which deal with cumulative summations of the variables. Now let us first we have to as I said we have to import the function from the for using that particular variable, particular function.

So in this case I am importing the cumulative sum which is coming from bouquet dot transform. So these transformations have to be informed, have to be imported from bouquet dot transform. I am importing the cumulative sum from that particular transformation function, I already have a source as column data source monthly values.

So the same thing I am using it here also. So then I am setting a figure with the same X range is equal to data source and then with the month names I am setting the figure, the figure is also set. Then again I am using a vertical bar for the passengers column alone.

Okay, so let us see what is happening here. So in this case I am plotting a vertical bar with month name January to December and the top, okay, which is the actual bar that is coming is cumulative sum of passengers, okay, include zero is true. Okay, this is how zero means you have to include the first element also in which the index is zero, width is 0.9. So this is for the bars and then you have the data source coming from the source column data transform.

Yes, running it and showing it, this is how it works. January month, February month, likewise, this is how the overall trend is going on. So you can clearly see that how each month and month and what is the total.

You also have the total at the end column, you can see by December how many passengers have travelled in that particular year. Okay, this is how you can use the transform functions to, you can apply transform functions on the existing data and then plot them very easily in the bokeh environment.

-----End-----